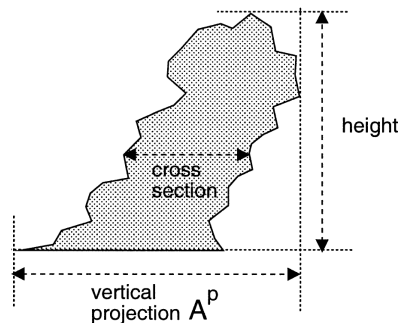


Size Statistics of Cumulus Cloud Populations in Large-Eddy Simulations



Three shallow-cumulus cases

BOMEX over the ocean.
SCMS and ARM over land.

A cloud is a connected set of saturated grid cells.
Its size is

$$l = \sqrt{A^p}$$

A^p : vertically projected area.

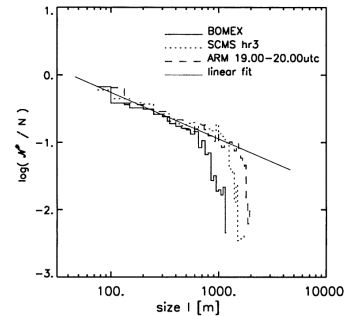
Neggers et al. (2003): measuring cloud size

The paper measures size statistics of shallow cumulus populations in large-eddy simulation (LES). The three reference cases are BOMEX, steady trade-wind cumulus over the ocean, SCMS over Florida, and the ARM diurnal cycle over Oklahoma. The reference horizontal grid spacing is 50 m, in a 6.4 km domain. Many snapshots give tens of thousands of sampled clouds per case.

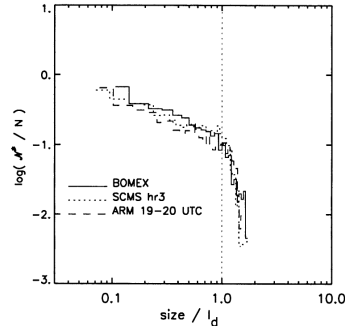
A cloud is a connected set of saturated grid boxes. Figure 3 distinguishes a cross-section at one altitude from the projection of the whole cloud onto the horizontal plane. The chosen linear size is the square root of that projected area, so it is neither a radius nor a diameter. This definition allows comparison with published size statistics from satellite images.

The paper then bins clouds by this size and measures their contributions to projected cover and vertical mass flux. The physical object throughout is a cumulus cloud. Source: Neggers et al. (2003), Sections 2–3 and Appendix, Figure 3, Equation (1).

The study measures cloud counts, cloud cover and mass flux by cloud size



Cloud size l



Rescaled size l/ℓ_d

Below the break

$$\mathcal{N}(l) \propto l^{-1.70}$$

About one decade of sizes.

Break scale ℓ_d

BOMEX: 700 m
SCMS: 1050 m
ARM: 400–1250 m

Rescaling by ℓ_d approximately collapses the three cloud-size densities

A power law with a case-dependent size break

The size density $\mathcal{N}(l)$ is the number of clouds per unit linear size. Below a case-dependent break, the fitted exponent is -1.70 over about one decade. The figures use logarithmic bins, $\mathcal{N}^* = l \ln(10) \mathcal{N}$, so their plotted slope is -0.70 . This distinction matters when reading the graph. Above the break the population falls off more steeply.

Figure 4a compares the cases at their original sizes. Figure 6 uses l/ℓ_d and shows an approximate collapse, supporting a common form for these simulated cloud populations. Table 2 gives 700 m for BOMEX, 1050 m for SCMS, and 400–1250 m through the ARM afternoon. The break is a cloud-size scale, not a distance between clouds.

The limited scaling range, three shallow-cumulus cases and unresolved physical origin of the break restrict generalization. The authors discuss subcloud-layer dynamics and shear as possible controls. No thermal-size or spacing distribution follows from this collapse. Source: Section 4a, Figures 4 and 6, Table 2, and Section 5.

Intermediate cloud sizes dominate vertical transport

3

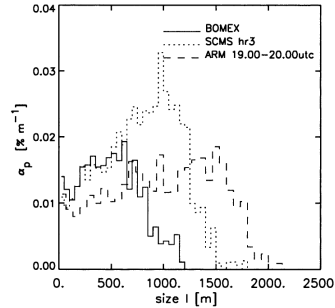
Paper review: Neggers et al. (2003) | Slide 3

Intermediate cloud sizes dominate vertical transport

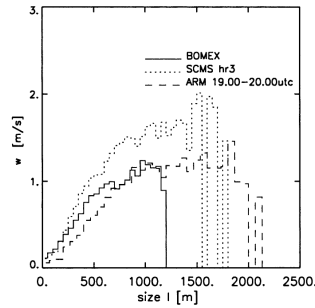
Read the three original panels from left to right. Projected cloud cover weights the size density by the projected area l^2 . It therefore has an intermediate maximum near the size break, even though small clouds are numerous. Cloud-mean vertical velocity increases with cloud size. The mass-flux decomposition, which combines cloud fraction and velocity and is averaged over the cloud layer, peaks at somewhat larger sizes than the projected-cover decomposition.

The result is a preferred range of cloud sizes for transport: the smallest clouds contribute very little to the total vertical mass flux. This conclusion concerns the plotted decompositions per unit size. The cover and flux peaks are related but are not exactly at the same size.

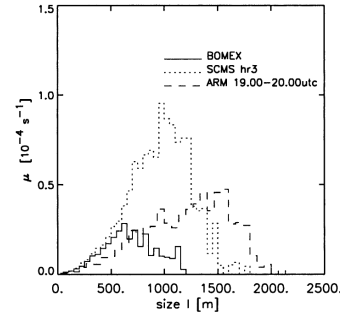
The dominant size is fairly insensitive to the tested grid spacings of 25–100 m. Too small a domain truncates the large clouds. Shear can increase projected cloud size without increasing total mass flux, which is another reason not to equate a projected cloud footprint with the section of its feeding updraft. Source: Section 4b–e, Figures 7, 9–11.



Cover peaks near the size break.



Vertical velocity grows with cloud size.



Mass flux peaks at somewhat larger sizes.

The smallest clouds contribute little to the total vertical mass transport

What is missing for a 2D thermal field

4

No distributions reported for

Thermal size at a specified height.

Thermal positions in the (x, y) plane.

Distances between thermals.

The cloud mask does not identify dry thermals below cloud base.

Possible use: our inference

The cloud-size break ℓ_d suggests a scale to compare with cloud-feeding updraft sizes.

Test this with paired cloud and updraft measurements at a fixed height.

Neither -1.70 nor ℓ_d transfers directly to a thermal-size law.

Cloud-size statistics offer a hypothesis to test, with no thermal spacing law

Scope: §3–5 and Appendix. The proposed comparison is our interpretation, not a result of Neggers et al. (2003).

Paper review: Neggers et al. (2003) | Slide 4

What is missing for a 2D thermal field

The purpose of our reading was to look for a statistical description of thermals in a horizontal plane. This paper does not report a thermal-size distribution at a fixed altitude, a distribution of thermal positions, a pair-correlation function or distances between thermal centres. It reports the projected sizes of saturated cloud objects. A projected area also contains cloud growth, decay, tilt and geometry over many heights.

A possible clue, explicitly our interpretation, is the finite cloud-size scale associated with the break and the dominant transport. The authors themselves discuss a possible connection to coherent structures in the subcloud layer, but do not establish it. A useful follow-up would identify buoyant updraft objects at a chosen height in LES and pair them with the clouds they feed. One could then test whether the cloud-size break has a counterpart in the updraft-size distribution. The definition of a thermal, the altitude and the velocity/buoyancy thresholds must be fixed.

Only under an additional, calibrated one-to-one relation such as $l_T = c l_C$ with constant c would a change of variables preserve the power-law exponent and rescale the break. The paper establishes neither this relation nor that all thermals form clouds. Cloudless thermals and multiple updrafts per cloud would invalidate that simple transfer. Even a valid size mapping would provide no spatial point process or spacing law. Source for the cloud statistics and open physical interpretation: Sections 3–5. All thermal extrapolations here are ours.